

表 5 不同自监督增强策略的独立作用

数据集	变体	AUROC _{max}	AP _{max}	Δ AUROC(%)	Δ AP(%)
TraceLog	no_ssl	0.8158	0.7291	+0.00%	+0.00%
	edge_perturbation	0.8084	0.7343	-0.74%	+0.52%
	edge_addition	0.8097	0.7309	-0.61%	+0.18%
	node_type_swap	0.8027	0.7284	-1.31%	-0.07%
	edge_type_swap	0.8131	0.7556	-0.27%	+2.66%
	topology_only	0.8079	0.7377	-0.79%	+0.87%
	type_only	0.8201	0.7614	+0.43%	+3.24%
	full	0.8208	0.7426	+0.50%	+1.35%
	all_four	0.8376	0.7778	+2.17%	+4.87%
FlowGraph	no_ssl	0.9996	0.9995	+0.00%	+0.00%
	edge_perturbation	0.9996	0.9995	+0.00%	+0.00%
	edge_addition	1.0000	1.0000	+0.04%	+0.05%
	node_type_swap	0.9708	0.9787	-2.88%	-2.08%
	edge_type_swap	0.9629	0.9757	-3.67%	-2.38%
	topology_only	0.9999	0.9999	+0.03%	+0.04%
	type_only	1.0000	1.0000	+0.04%	+0.05%
	full	0.9828	0.9848	-1.68%	-1.47%
	all_four	1.0000	1.0000	+0.04%	+0.05%
ADFA-LD	no_ssl	0.8269	0.8260	+0.00%	+0.00%
	edge_perturbation	0.8443	0.8383	+1.74%	+1.23%
	edge_addition	0.8515	0.8358	+2.45%	+0.98%
	node_type_swap	0.8302	0.8245	+0.32%	-0.15%
	edge_type_swap	0.8316	0.8250	+0.46%	-0.10%
	topology_only	0.8584	0.8444	+3.14%	+1.84%
	type_only	0.8326	0.8259	+0.56%	-0.01%
	full	0.8474	0.8421	+2.05%	+1.61%
	all_four	0.8650	0.8526	+3.80%	+2.66%

说明： Δ AUROC 和 Δ AP 均以同一数据集上的 no_ssl 为参照，单位为百分点。表中各变体按该变体已完成运行的最大 AUROC/AP 报告；all_four 行直接采用主表 HRA-GNN 最佳运行，以保证完整方法口径与主表一致。